FACULTY OF COMPUTING & INFORMATICS

MULTIMEDIA UNIVERSITY

Final Year Project Proposal Form

Project Title	Development of a Machine Learning-Based System for Cardiopulmonary Sound Separation
Supervisor Name	Mr. Jamal Arshad
Co-Supervisor Name (if any)	
Project Status	Student-Proposed
Project Type	Application-Based
Project Specialization (Project Specialisation and Student Specialisation should match)	Software Engineering
Project Category (Pls. refer at the end of document for the selection of category based on the specialisation)	Application Software
Project Focus/Contribution (Pls. refer at the end of document for the selection of focus/contribution based on the specialisation)	Prototype/Proof of Concept

Project Description **(Discuss Background, Problem Statement, Methodology, Expected Output/Significance in summary form)**	**Background:** Cardiopulmonary sounds, including heart and lung sounds, are commonly recorded together using a stethoscope. These sounds often overlap, making it difficult to separate them using traditional signal processing methods. **Problem Statement:** Existing approaches struggle to accurately separate heart and lung sounds due to overlapping frequencies and noise. Additionally, most existing solutions focus on medical diagnosis rather than developing a structured and reusable software system for sound separation. **Methodology:** The proposed system will be developed using Python and machine learning techniques. The system will include audio preprocessing, feature extraction, and implementation of machine learning models to separate heart and lung sounds from mixed audio signals. Public datasets will be used for training and testing. **Expected Output / Significance:** The system will produce two separate audio outputs (heart and lung sounds) from a mixed input signal. It will provide a functional prototype demonstrating sound separation and contribute as a reusable software solution for future medical informatics research.

Project Objectives/ Outcomes **(Focused and precise list of statements that can imply the goals to be achieved, Majority of the Project Objectives/Outcomes are in line with the Project and Student specialisation)**	1. To study existing techniques for cardiopulmonary sound separation. 2. To design and develop a machine learning-based system for sound separation. 3. To implement and test the system using real audio datasets. 4. To evaluate the performance of the system using suitable metrics.
Project Scope (Focus/Expected Output/ Deliverables with the limits and constraints of the study can be described and implies enough scope for the two-trimester project)	**Expected Outputs:** 1. A software system for separating heart and lung sounds. 2. Implementation of preprocessing and machine learning modules. 3. Output of separated audio signals. **Limitations and Constraints:** 1. Limited to sound separation only (no disease detection). 2. Dependence on publicly available datasets. 3. Limited timeframe (two trimesters). 4. Performance depends on model accuracy.

Number of Students (If it is two-students project, subtitles and work distribution must be clearly specified and differentiated for each student)	One
Student 1 Subtitle (Pls. fill up if the number of students is two)	
Student 1 Work Distribution (Pls. fill up if the number of students is two)	
Student 2 Subtitle (Pls. fill up if the number of students is two)	
Student 2 Work Distribution (Pls. fill up if the number of students is two)	
Industry Collaboration	No
Company Name, Contact Name, Contact Phone (if the answer to Industry Collaboration is Yes)	

Student 1 Details (Student Name, Student ID, Specialisation, Handphone number, E-mail address) **(Leave it blank, if unknown)**	**Name:** AL-SALOUL, ASHRAF ALI HUSSEIN **ID:** 1221303805 **Specialization:** Software Engineering **Phone Number:** 0183816023 **Pesonal E-mail address:** thalththanwyd@gmail.com **MMU E-mail address:** 1221303805@student.mmu.edu.my
Student 2 Details (if it is a two-student project) (Student Name, Student ID, Specialisation, Handphone Number, E-mail address) **(Leave it blank, if unknown)**	

Select one Project Category based on Specialisation:

Software Engineering:

Critical System

Application Software

Software Tools & Utilities

Service Oriented Computing

Data Science:

Data Engineering

Data Analytics

Cybersecurity:

Cryptography and Data Security

Investigation and Analysis

Security and Defence

Game Development:

Game Software Development (GSD)

Game Algorithm Research (GAR)

Game Design Prototyping (GDP)

Information Systems:

IT Infrastructure

Transaction Processing Systems

Intelligent Systems

Select one Focus/Contribution based on Specialisation:

Software Engineering:

Product Development

Prototype/Proof of Concept

Software Engineering Methodologies

Others (Pls. specify)

Data Science:

Data Management

IoT

Optimisation of Technologies

Analysis of data (texts, videos, images, numerical digit)

Others (Pls. specify)

Cybersecurity:

Cryptography

Database Security

Blockchain

Malware analysis

Forensics

Ethical hacking

Network and Cloud Security

Others (Pls. specify)

Game Development:

Game Software Development (GSD): Development and implementation of a complete game from design, programming to production of a complete game installation package

Game Algorithm Research (GAR): Thorough investigation and analysis of specific algorithms used in games

Game Design Prototyping (GDP): Proof of concept of novel specific game design concepts or game mechanics via development of complete prototypes

Information Systems:

Data & Information Management

User Experience

System Analysis & Design

IS Project Management

Business Processes

Technology Evaluation

Others (Pls. specify)